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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Sc DEGREE EXAMINATION APRIL 2021

ASMT202S - MATHEMATICAL STATISTICS - II

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2 = 20)

Answer ALL the questions

1. Obtain the moment generating function of the Poisson distribution.

2. Give the density function of Geometric distribution.

3. If X is uniformly distributed with mean 1 and variance 4/3, find P(X < 0).

4. What is the cumulative distribution function of exponential distribution?

5. State any two applications of t-distribution in testing of significance.

6. Define degrees of freedom.

7. What is the test statistic for testing the significance for single proportion?

8. Define Chi-Square test.

9. State the assumptions in analysis of variance.

10. State any two advantages of CRD.

SECTION – B (5X5=25)

Answer **ALL** the Questions

11. (a) Ten coins are thrown simultaneously. Find the probability of getting atleast seven heads.

(Or)

(b) Obtain mean and variance of Geometric distribution.

12. (a) Obtain density function of student's t – distribution with n degrees of freedom.

(Or)

(b) The mean yield for one acre plot is 662 kilos with standard deviation 32 kilos. Assuming normal distribution , how many one acre plots in a batch of 1000 plots would you expect to have yield (i) over 700 kilos (ii) below 650 kilos.

13 (a) How will you test paired t-test for difference of means in the case of large samples?

(Or)

(b) For the 2 X 2 table,

a	b
c	d

Obtain Chi-square test of independence.

14. (a) Explain the procedure for testing of significance for difference in means for large samples.

(Or)

(b) How will you test the significance of the difference between two independent sample correlation coefficients.

15 (a) Describe ANOVA table for one-way classification model.

(Or)

(b) Explain the steps in constructing Latin Square Design.

SECTION - C (3X10=30)

Answer any **THREE** Questions

16. Derive probability density function of Poisson distribution.

17. Obtain the probability density function of chi-square distribution with n degrees of freedom.

18. Samples of two types of electric light bulbs were tested for length of life and following data were obtained.

	TYPE I	TYPE II
Sample number	$n_1 = 8$	$n_2 = 7$
Sample means	$\overline{x_1} = 1234$ hrs	$\overline{x_2} = 1036$ hrs
Sample standard deviations	$S_1 = 36$ hrs	$S_2 = 40$ hrs

Is the difference in means sufficient to warrant that TYPE I is superior to TYPE II regarding the length of life?

19. In one sample of 8 observations, the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in the other sample of 10 observations it was 102.6. Test whether this difference is significant at 5% level given that 5% of F for $n_1 = 7$ and $n_2 = 9$ is 3.29.

20. Explain Randomized Block design and prepare its structure of analysis of variance table.

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